In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")
```

In [2]:
```
matches = pd.read_csv("matches.csv")
deliveries = pd.read_csv("deliveries.csv")

matches.head()
```

Out[2]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	r
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	no
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	no
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	no
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	no
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	no

In [3]:
```
matches = pd.read_csv("matches.csv")
deliveries = pd.read_csv("deliveries.csv")

matches.head()
```

Out[3]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	r
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	no
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	no
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	no
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	no
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	no

In [4]:

```
matches.info()
matches.describe()
matches.isnull().sum()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 756 entries, 0 to 755
Data columns (total 18 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   id               756 non-null    int64
 1   season           756 non-null    int64
 2   city             749 non-null    str
 3   date             756 non-null    str
 4   team1            756 non-null    str
 5   team2            756 non-null    str
 6   toss_winner      756 non-null    str
 7   toss_decision    756 non-null    str
 8   result           756 non-null    str
 9   dl_applied       756 non-null    int64
 10  winner           752 non-null    str
 11  win_by_runs      756 non-null    int64
 12  win_by_wickets   756 non-null    int64
 13  player_of_match  752 non-null    str
 14  venue            756 non-null    str
 15  umpire1          754 non-null    str
 16  umpire2          754 non-null    str
 17  umpire3          119 non-null    str
dtypes: int64(5), str(13)
memory usage: 106.4 KB
```

Out[4]:
```
id                   0
season               0
city                 7
date                 0
team1                0
team2                0
toss_winner          0
toss_decision        0
result               0
dl_applied           0
winner               4
win_by_runs          0
win_by_wickets       0
player_of_match      4
venue                0
umpire1              2
umpire2              2
umpire3            637
dtype: int64
```

In [5]:
```
matches.info()
matches.describe()
matches.isnull().sum()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 756 entries, 0 to 755
Data columns (total 18 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   id               756 non-null    int64
 1   season           756 non-null    int64
 2   city             749 non-null    str
 3   date             756 non-null    str
 4   team1            756 non-null    str
 5   team2            756 non-null    str
 6   toss_winner      756 non-null    str
 7   toss_decision    756 non-null    str
 8   result           756 non-null    str
 9   dl_applied       756 non-null    int64
 10  winner           752 non-null    str
 11  win_by_runs      756 non-null    int64
 12  win_by_wickets   756 non-null    int64
 13  player_of_match  752 non-null    str
 14  venue            756 non-null    str
 15  umpire1          754 non-null    str
 16  umpire2          754 non-null    str
 17  umpire3          119 non-null    str
dtypes: int64(5), str(13)
memory usage: 106.4 KB
```

Out[5]:
```
id                   0
season               0
city                 7
date                 0
team1                0
team2                0
toss_winner          0
toss_decision        0
result               0
dl_applied           0
winner               4
win_by_runs          0
win_by_wickets       0
player_of_match      4
venue                0
umpire1              2
umpire2              2
umpire3            637
dtype: int64
```

In [6]:
```
matches.info()
matches.describe()
matches.isnull().sum()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 756 entries, 0 to 755
Data columns (total 18 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   id               756 non-null    int64
 1   season           756 non-null    int64
 2   city             749 non-null    str
 3   date             756 non-null    str
 4   team1            756 non-null    str
 5   team2            756 non-null    str
 6   toss_winner      756 non-null    str
 7   toss_decision    756 non-null    str
 8   result           756 non-null    str
 9   dl_applied       756 non-null    int64
 10  winner           752 non-null    str
 11  win_by_runs      756 non-null    int64
 12  win_by_wickets   756 non-null    int64
 13  player_of_match  752 non-null    str
 14  venue            756 non-null    str
 15  umpire1          754 non-null    str
 16  umpire2          754 non-null    str
 17  umpire3          119 non-null    str
dtypes: int64(5), str(13)
memory usage: 106.4 KB
```

Out[6]:
```
id                   0
season               0
city                 7
date                 0
team1                0
team2                0
toss_winner          0
toss_decision        0
result               0
dl_applied           0
winner               4
win_by_runs          0
win_by_wickets       0
player_of_match      4
venue                0
umpire1              2
umpire2              2
umpire3            637
dtype: int64
```

In [7]:
```
max_runs = matches.loc[matches['win_by_runs'].idxmax()]
max_runs
```

Out[7]:
```
id                               44
season                         2017
city                          Delhi
date                     2017-05-06
team1                Mumbai Indians
team2               Delhi Daredevils
toss_winner         Delhi Daredevils
toss_decision                 field
result                       normal
dl_applied                        0
winner               Mumbai Indians
win_by_runs                     146
win_by_wickets                    0
player_of_match         LMP Simmons
venue               Feroz Shah Kotla
umpire1                 Nitin Menon
umpire2                   CK Nandan
umpire3                         NaN
Name: 43, dtype: object
```

In [8]:
```
max_wickets = matches.loc[matches['win_by_wickets'].idxmax()]
max_wickets
```

Out[8]:
```
id                                                   3
season                                            2017
city                                             Rajkot
date                                         2017-04-07
team1                                     Gujarat Lions
team2                               Kolkata Knight Riders
toss_winner                         Kolkata Knight Riders
toss_decision                                       field
result                                             normal
dl_applied                                              0
winner                              Kolkata Knight Riders
win_by_runs                                             0
win_by_wickets                                         10
player_of_match                                   CA Lynn
venue              Saurashtra Cricket Association Stadium
umpire1                                        Nitin Menon
umpire2                                          CK Nandan
umpire3                                                NaN
Name: 2, dtype: object
```

In [9]:
```
min_runs = matches[matches['win_by_runs'] > 0].sort_values('win_by_runs').head(1
min_runs
```

Out[9]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	res
58	59	2017	Hyderabad	2017-05-21	Mumbai Indians	Rising Pune Supergiant	Mumbai Indians	bat	norr

In [10]:
```
min_wickets = matches[matches['win_by_wickets'] > 0].sort_values('win_by_wickets
min_wickets
```

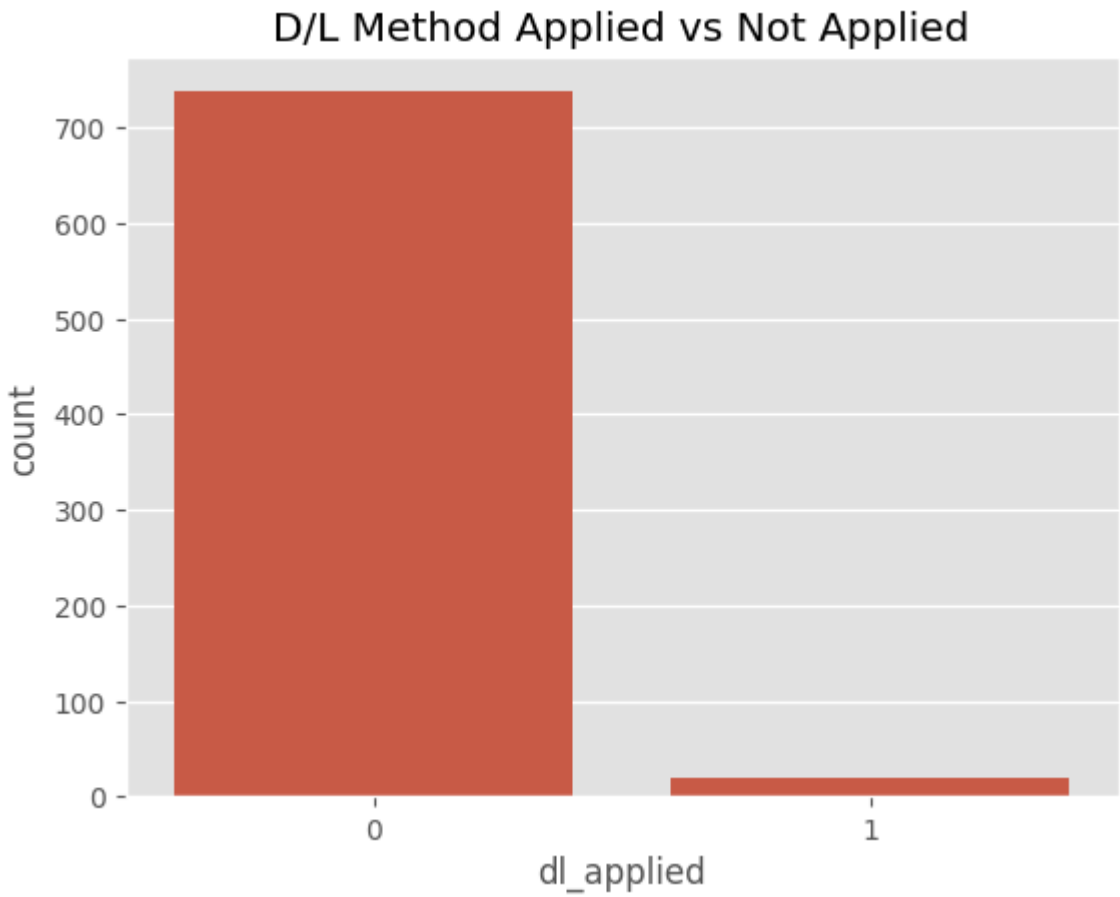
Out[10]:

	id	season	city	date	team1	team2	toss_winner	toss_decisior
642	7900	2018	Hyderabad	12/04/18	Mumbai Indians	Sunrisers Hyderabad	Sunrisers Hyderabad	fielc

In [11]:
```
matches['dl_applied'].value_counts()
```

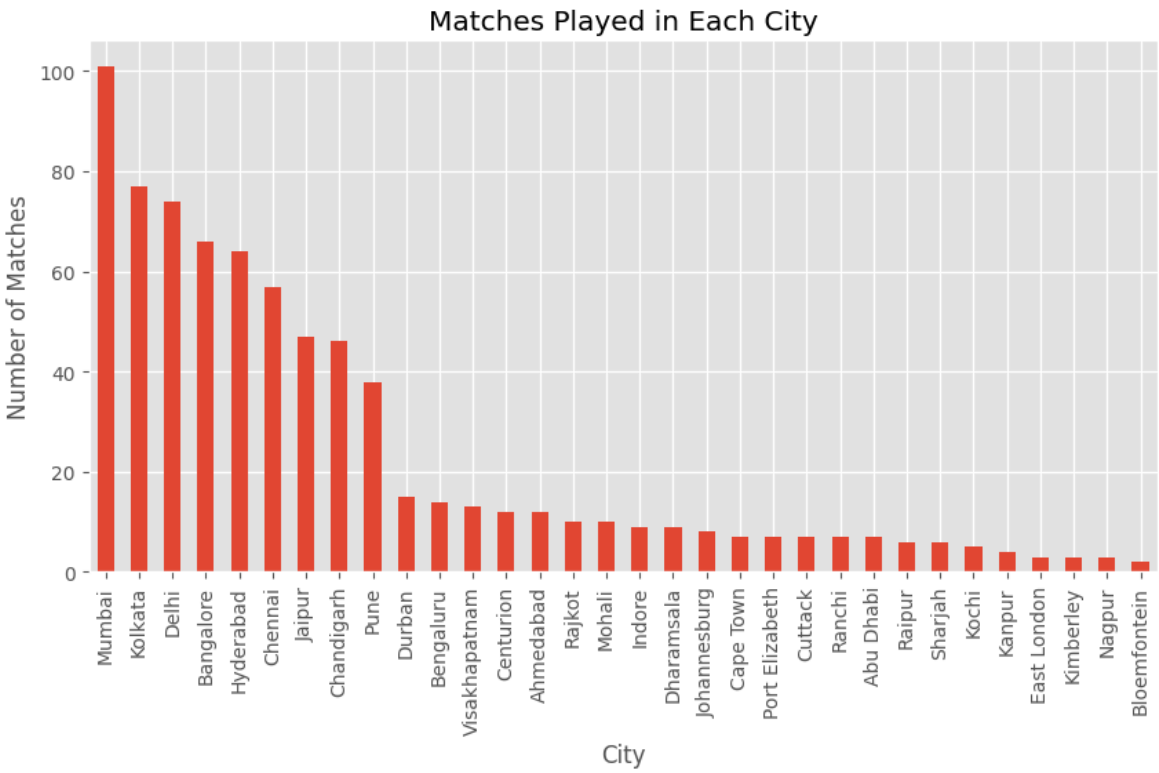
Out[11]:
```
dl_applied
0    737
1     19
Name: count, dtype: int64
```

In [12]:
```
sns.countplot(x='dl_applied', data=matches)
plt.title("D/L Method Applied vs Not Applied")
plt.show()
```



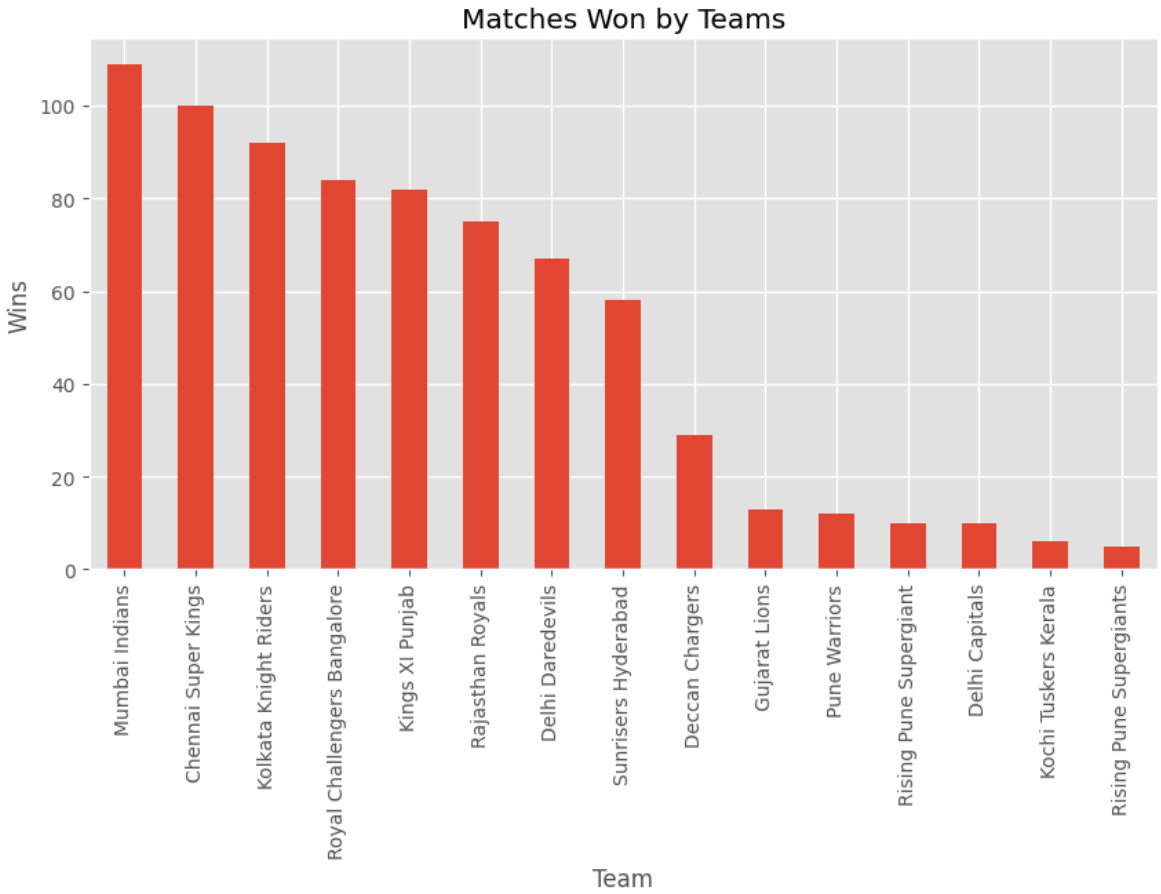
In [14]:
```
city_matches = matches['city'].value_counts()

city_matches.plot(kind='bar', figsize=(10,5))
plt.title("Matches Played in Each City")
plt.xlabel("City")
plt.ylabel("Number of Matches")
plt.show()
```



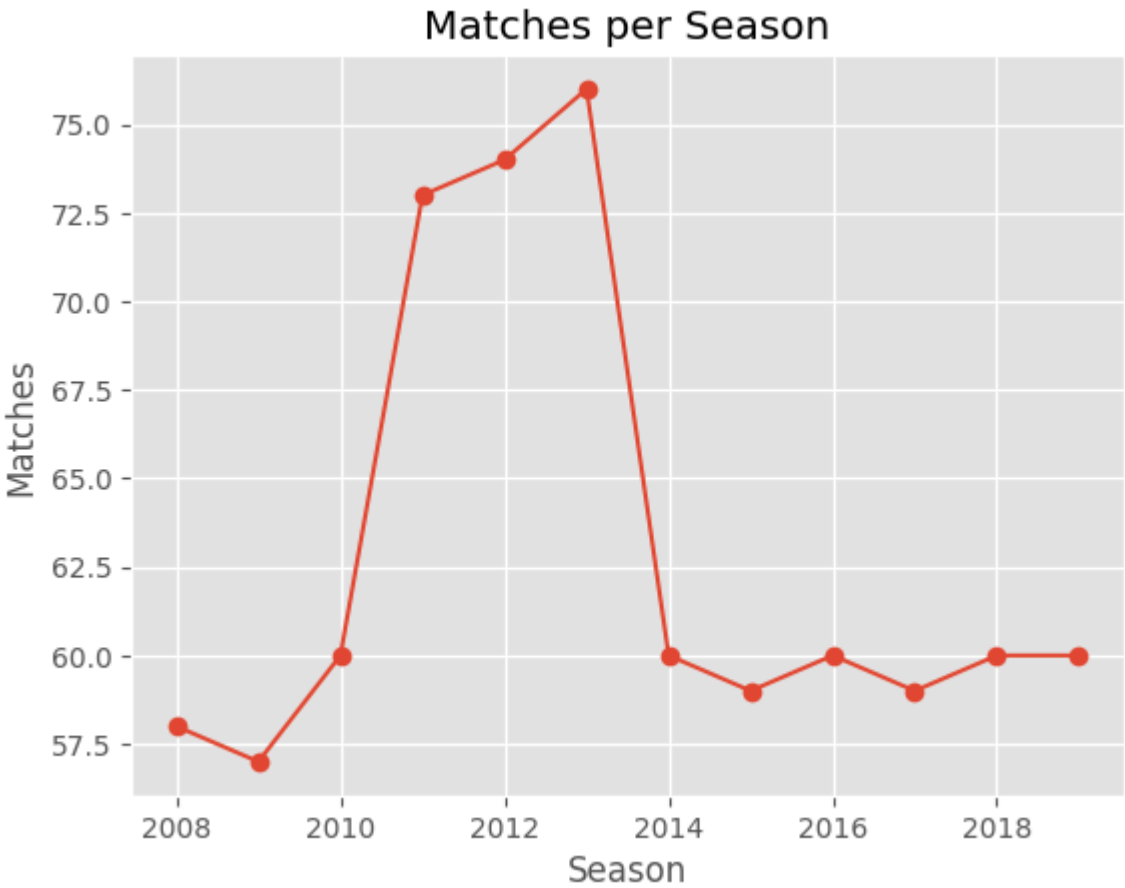
In [15]:
```
team_wins = matches['winner'].value_counts()

team_wins.plot(kind='bar', figsize=(10,5))
plt.title("Matches Won by Teams")
plt.xlabel("Team")
plt.ylabel("Wins")
plt.show()
```



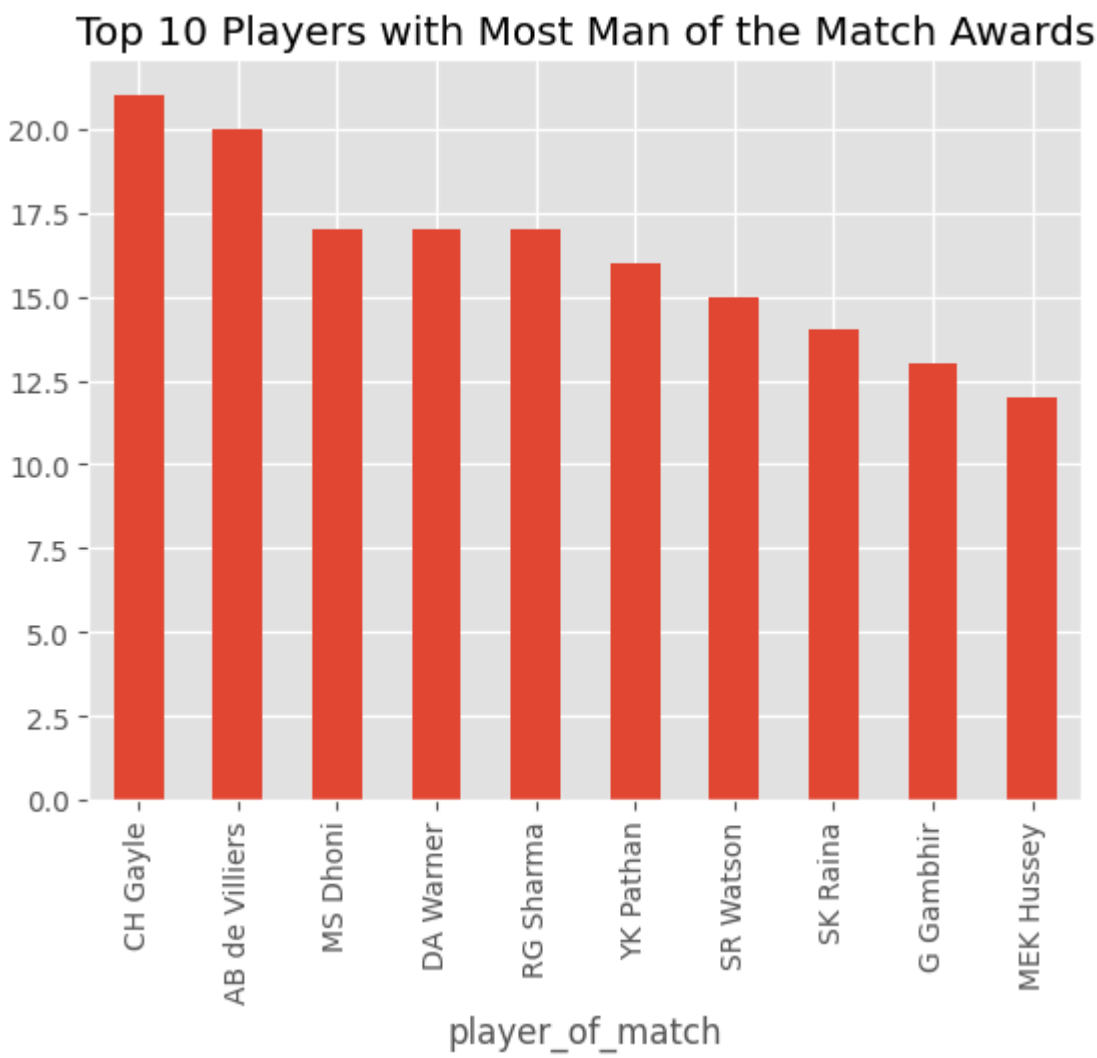
In [16]:
```
season_matches = matches['season'].value_counts().sort_index()

season_matches.plot(kind='line', marker='o')
plt.title("Matches per Season")
plt.xlabel("Season")
plt.ylabel("Matches")
plt.show()
```



In [17]:
```
mom = matches['player_of_match'].value_counts().head(10)

mom.plot(kind='bar')
plt.title("Top 10 Players with Most Man of the Match Awards")
plt.show()
```

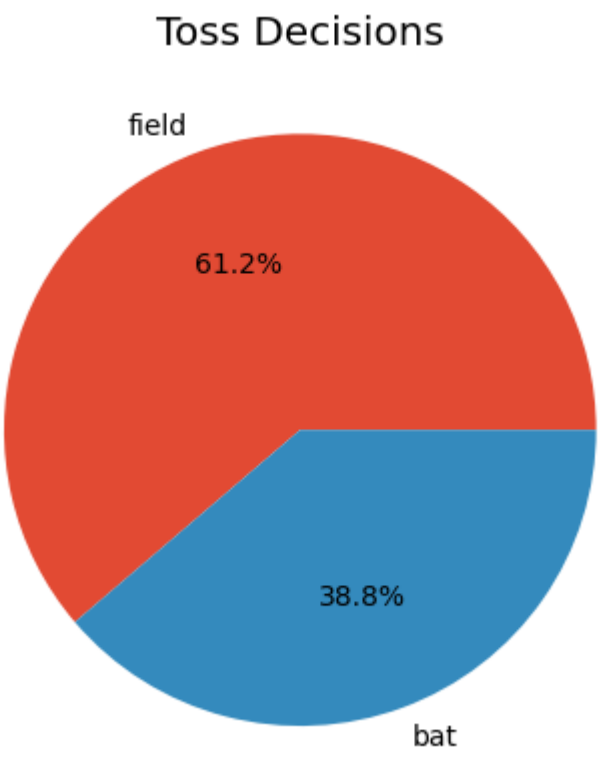


In [18]:
```
toss_match = matches[matches['toss_winner'] == matches['winner']]
percentage = (len(toss_match) / len(matches)) * 100
percentage
```

Out[18]: 51.98412698412699

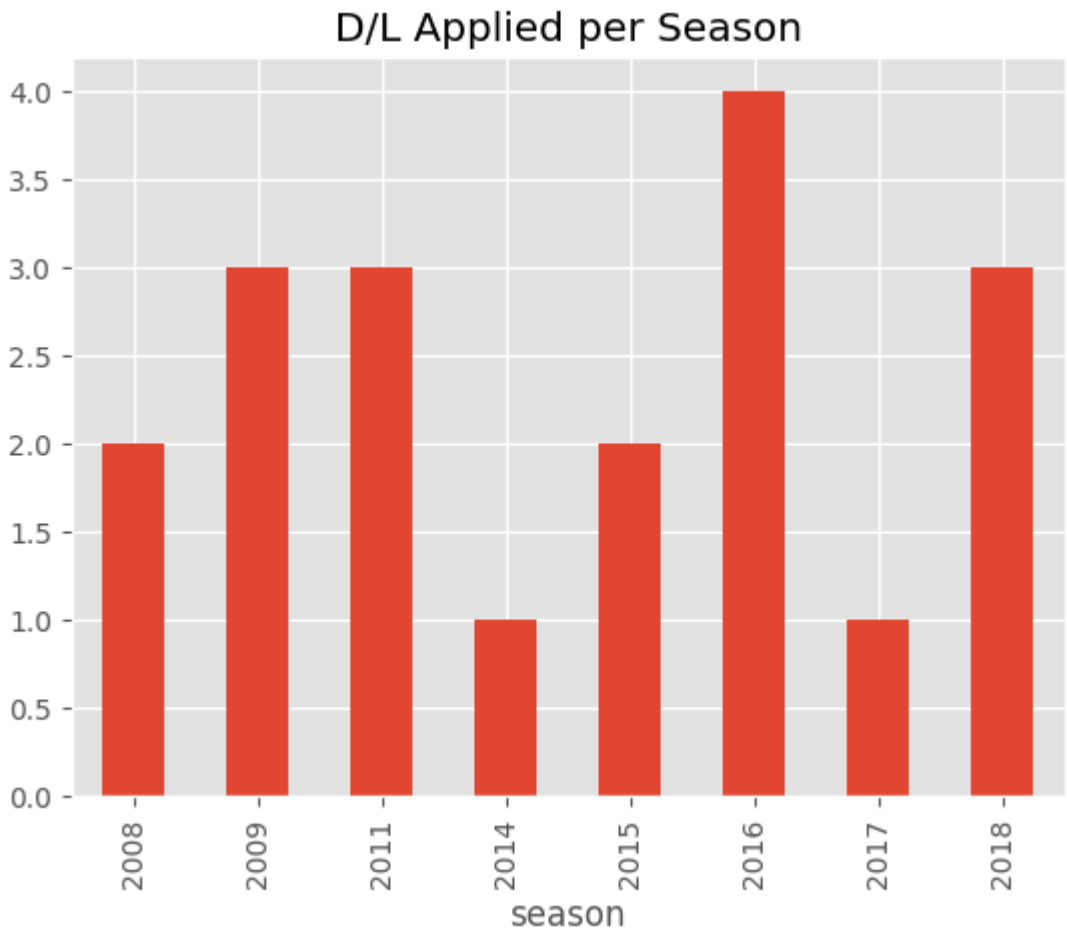
In [19]:
```
toss_decision = matches['toss_decision'].value_counts()

toss_decision.plot(kind='pie', autopct='%1.1f%%')
plt.title("Toss Decisions")
plt.show()
```



In [20]:
```
dl_season = matches[matches['dl_applied']==1].groupby('season').size()

dl_season.plot(kind='bar')
plt.title("D/L Applied per Season")
plt.show()
```

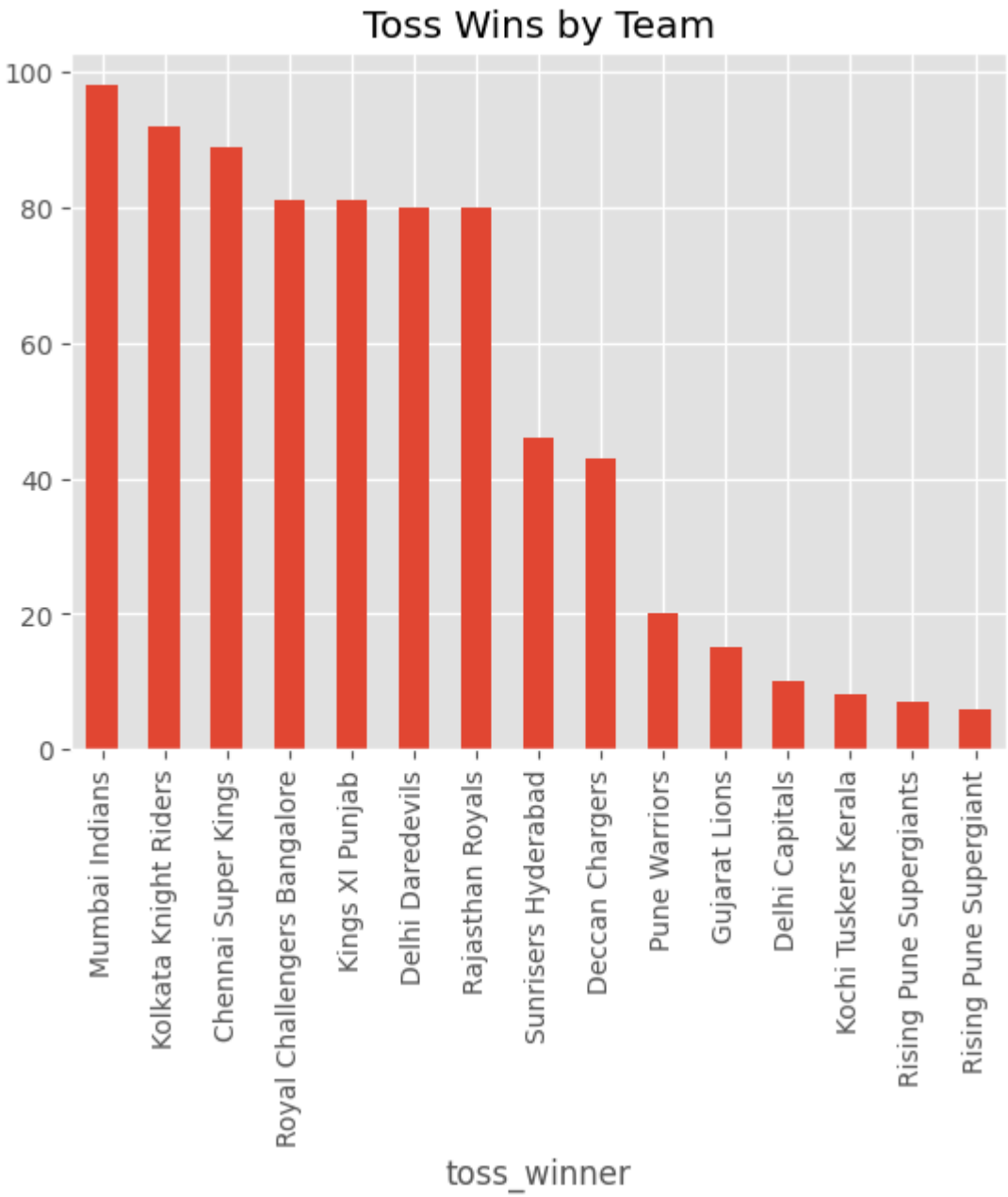


In [21]:
```
matches['result'].value_counts()
```

Out[21]:
```
result
normal       743
tie            9
no result      4
Name: count, dtype: int64
```

In [22]:
```
toss_wins = matches['toss_winner'].value_counts()

toss_wins.plot(kind='bar')
plt.title("Toss Wins by Team")
plt.show()
```



In [23]:
```
defending = matches[matches['win_by_runs'] > 0]
defending['venue'].value_counts().head(5)
```

Out[23]:
```
venue
Wankhede Stadium                    35
Feroz Shah Kotla                    33
Eden Gardens                        32
M Chinnaswamy Stadium               30
MA Chidambaram Stadium, Chepauk     30
Name: count, dtype: int64
```

In [24]:
```
chasing = matches[matches['win_by_wickets'] > 0]
chasing['venue'].value_counts().head(5)
```

Out[24]:
```
venue
Eden Gardens                                45
M Chinnaswamy Stadium                       40
Wankhede Stadium                            37
Feroz Shah Kotla                            33
Rajiv Gandhi International Stadium, Uppal   32
Name: count, dtype: int64
```

In [25]:
```
defend_team = defending['winner'].value_counts().head(5)
defend_team
```

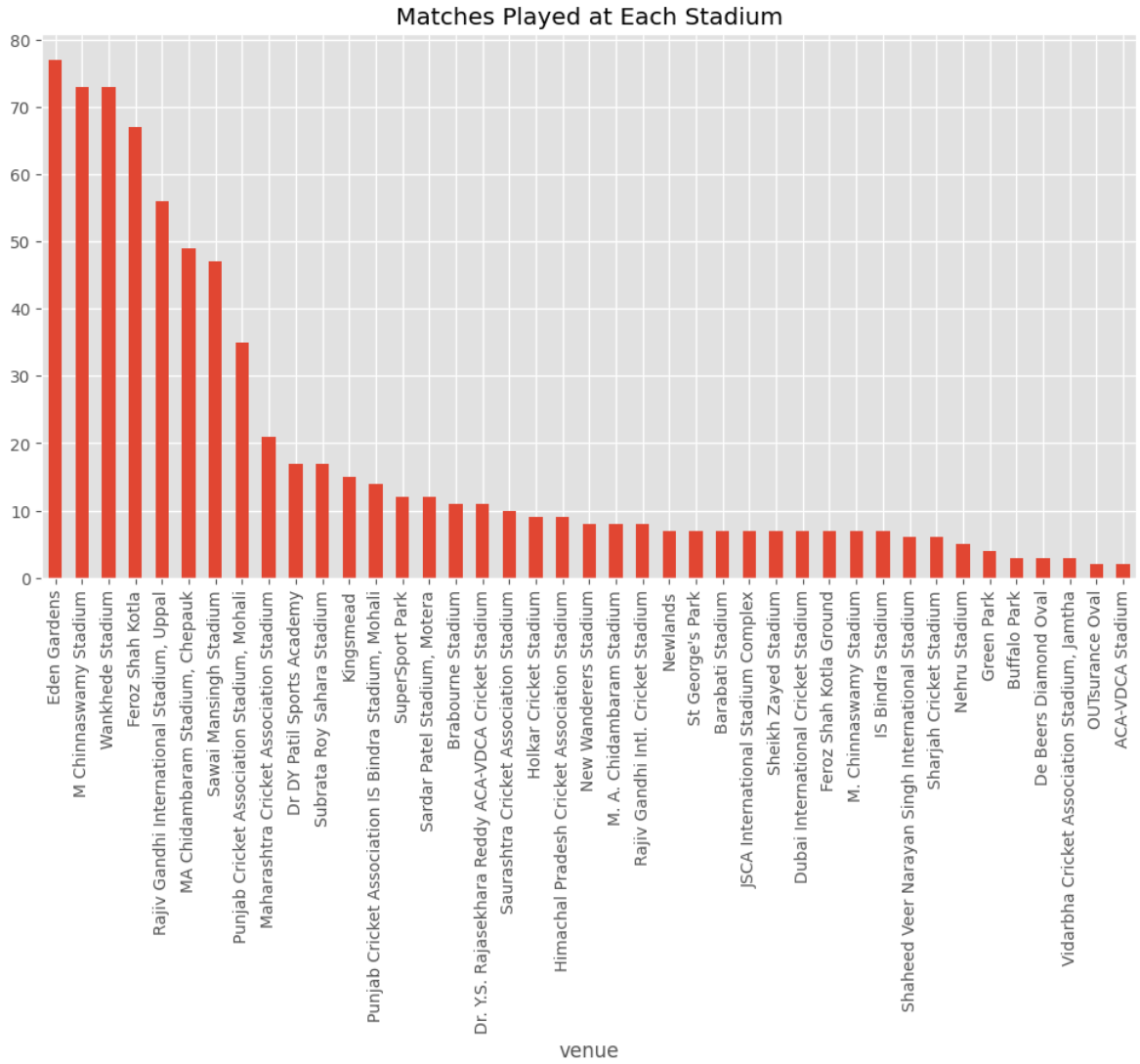
Out[25]:
```
winner
Mumbai Indians                  57
Chennai Super Kings             52
Kings XI Punjab                 38
Kolkata Knight Riders           36
Royal Challengers Bangalore     35
Name: count, dtype: int64
```

In [26]:
```
chase_team = chasing['winner'].value_counts().head(5)
chase_team
```

Out[26]:
```
winner
Kolkata Knight Riders           56
Mumbai Indians                  50
Royal Challengers Bangalore     48
Chennai Super Kings             48
Rajasthan Royals                46
Name: count, dtype: int64
```

In [27]:
```
venue_matches = matches['venue'].value_counts()

venue_matches.plot(kind='bar', figsize=(12,6))
plt.title("Matches Played at Each Stadium")
plt.show()
```


Matches Played at Each Stadium
80
70
60
50
40
30
20
10
0
Eden Gardens
M Chinnaswamy Stadium
Wankhede Stadium
Feroz Shah Kotla
Rajiv Gandhi International Stadium, Uppal
MA Chidambaram Stadium, Chepauk
Sawai Mansingh Stadium
Punjab Cricket Association Stadium, Mohali
Maharashtra Cricket Association Stadium
Dr DY Patil Sports Academy
Subrata Roy Sahara Stadium
Kingsmead
Punjab Cricket Association IS Bindra Stadium, Mohali
SuperSport Park
Sardar Patel Stadium, Motera
Brabourne Stadium
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium
Saurashtra Cricket Association Stadium
Holkar Cricket Stadium
Himachal Pradesh Cricket Association Stadium
New Wanderers Stadium
M. A. Chidambaram Stadium
Rajiv Gandhi Intl. Cricket Stadium
Newlands
St George's Park
Barabati Stadium
JSCA International Stadium Complex
Sheikh Zayed Stadium
Dubai International Cricket Stadium
Feroz Shah Kotla Ground
M. Chinnaswamy Stadium
IS Bindra Stadium
Shaheed Veer Narayan Singh International Stadium
Sharjah Cricket Stadium
Nehru Stadium
Green Park
Buffalo Park
De Beers Diamond Oval
Vidarbha Cricket Association Stadium, Jamtha
OUTsurance Oval
ACA-VDCA Stadium
venue

In [ ]: